

Predicting Hit Songs Using Repeated Chorus

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Research question

Can audio measurements from a 15-second chorus distinguish a Billboard year-end hit from another song that appeared on the weekly Hot 100? Both classes already charted. The study tests a retrospective association, not a formula for making a hit.

Dataset and method

We use an inherited table of 751 songs, 71 normalized artist-name groups and 518 audio measurements. The original development set contains 597 songs from 52 groups; 154 songs from 19 other groups form an already known historical test. Artist names and titles are excluded from model inputs.

The current study compared 63 declared settings using five outer and three inner artist-grouped folds. Eight new settings used 74 features measuring variation in chorus audio. All learned preprocessing and model selection stayed within training folds. The final development model is balanced logistic regression with standard scaling and 74 selected features.

Measured development results

Metric	Previous V3	Current V4
Accuracy	53.27%	54.61%
Balanced accuracy	53.25%	54.66%
Precision (year-end hit)	51.86%	53.07%
Recall (year-end hit)	52.76%	56.55%
F1 score	52.31%	54.76%

V4 balanced accuracy is **54.66%** (approximate 95% whole-artist interval 50.72%-58.50%). Its paired change from V3 is +1.41 percentage points (interval -4.29 to +7.33 points). The interval includes zero, so a dependable gain is not established.

Conclusion

The software, grouped evaluation and demo are complete, but the observed accuracy remains modest. No fresh independent test exists, and the original matching recordings are unavailable for verifying audio extraction. The immediate research need is better recording/label evidence and a new evaluation collection.

Code and evidence: github.com/DhanushSS/chorus-hit-predictor. Source features: [Antonios Malak dataset](#) (Apache-2.0).